

```
$ sudo apt-get install build-essential protobuf-compiler
python \
    libprotobuf-dev libcurl4-openssl-dev \
    libboost-all-dev libncurses5-dev \
    libjemalloc-dev wget m4
$ wget https://download.rethinkdb.com/dist/rethinkdb-version.
tgz
$ tar xf rethinkdb-version.tgz
$ cd rethinkdb-version
$ ./configure --allow-fetch
$ make
$ sudo make install
```

To start the server on Ubuntu Linux, execute the following instruction from the terminal window:

```
$ rethinkdb
```

Creating a table using the Data Explorer tab in Explorer

In the RethinkDB administrative console, there is a Data Explorer tab, which provides the panel for executing the commands that will work on the databases. In the following example, a new table titled *chat_users* is created in the database titled *mydatabase*.

```
r.db('mydatabase').tableCreate('chat_users')
```

On clicking the *Run* button at the bottom-right of the administration console, the query is executed and the operation is implemented. In addition, the key combination *Shift+Enter* can be used to run the query entered.

To insert records in the form of JSON in the table *chat_users*, use the following commands:

```
r.table('chat_users').insert([
  { name: 'Username-1', age: 17 },
  { name: 'Username-2', age: 35 }])
```

To count the number of records in the table, give the following command:

```
r.table('chat_users').count()
```

To display the records of application users aged 20 years and above from the table, use the following command:

```
r.table('chat_users').filter(r.row('age').gt(20))
```

Programming with RethinkDB using Python

Python is a free and open source programming language that has interfaces for almost all NoSQL databases. It is widely used for cloud applications and assorted real-time high-performance Web applications. In Python, RethinkDB can be easily installed and mapped using Pip, using the following

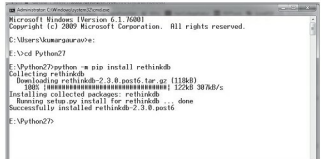


Figure 8: Installation of RethinkDB with Python

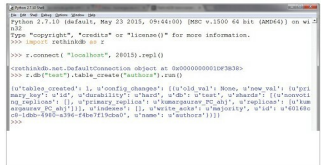


Figure 9: Execution of RethinkDB instructions in the Python IDLE shell

command:

```
WindowsDirectory:> python -m pip install rethinkdb
```

The package installer of Python automatically fetches the libraries of RethinkDB and maps them with the existing installation of Python.

After mapping RethinkDB with Python, the IDLE shell can use the methods and APIs of RethinkDB.

```
>> import rethinkdb as <Handle>
```

Using the Python IDLE shell, or within the script, the connection can be opened with the following command, which includes a client driver with 28015 as the port number:

```
>> r.connect("localhost", 28015).repl()
```

The *repl* command sets and prepares the default connection with the Python shell.

The function *table_create()* is used to add a new table in the particular database, as shown in Figure 9. 

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